Sahaj Bhadja

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Summary

Software engineer with 2+ years building data-backed applications, ETL workflows, REST APIs, and analytics-ready systems. Strong in Python, SQL, cloud pipelines, testing, and reliable user-facing insights.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
M.S. in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Web Tech, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
B.Tech. in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Eng., OS, OOP, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed SQL-backed platform workflows for 500+ users, tuning indexed queries, validation logic, and background jobs across FastAPI services to cut dashboard latency from 1.2s to 350ms.
- Built ETL-style integration pipelines using SQL Server, PostgreSQL, and cloud storage, processing 5,000+ monthly jobs with reconciliation checks, error handling, and reliable CI/CD-backed releases.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL integrations with Python, SQL, AWS Glue, and S3, improving response time by 30% while supporting scalable source-to-target workflows and maintainable backend components.
- Maintained 20+ automated tests and Jenkins pipelines with PyTest and Mockito, increasing CI reliability by 68% and strengthening validation, deployment, and debugging discipline across shared environments.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Flask services, REST APIs, and React workflows handling 10,000+ daily requests, integrating secure uploads, grading history, and analytics-ready product flows for a high-volume web application.
- Designed concurrent batch-processing for 1,000+ uploaded files using multithreading, AWS RDS, and CI/CD-backed releases, reducing feedback turnaround by 60% while maintaining stable APIs under peak demand.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Created Python evaluation pipelines across 100,000+ records, producing structured metrics, logs, and visual summaries that supported repeatable model analysis and data-quality review.
- Documented prompt, evaluation, and reporting workflows for researchers, turning ad hoc experiments into reproducible processes with clearer traceability, faster handoffs, and more reliable reporting.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built full-stack product features with REST APIs, WebSockets, and responsive interfaces, supporting 2M+ synchronized records across modular workflows and reliable end-to-end application behavior.
- Automated Docker-based build and test flows with Jenkins CI/CD, cutting release time by 50% and improving deployment consistency, troubleshooting speed, and engineering collaboration across product environments.

Skills

Languages: Python, SQL, JavaScript, TypeScript, Java, C++, Bash
Data Engineering: ETL/ELT, data modeling, source-to-target mapping, data quality, reconciliation, AWS Glue, S3, SQL Server, PostgreSQL, MySQL, MongoDB
Platforms & APIs: Flask, FastAPI, Node.js, Express, REST APIs, JSON, background jobs, cloud storage
Analytics & Delivery: Databricks exposure, BI-style dashboards, structured metrics, Git, Jenkins, CI/CD, Docker, Agile/Scrum
Engineering: unit testing, debugging, code reviews, naming conventions, documentation, scalability, ownership

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*
- Built an AI-powered SQL workspace supporting 10+ databases, accelerating query authoring, schema exploration, and structured data management through a unified Java and TypeScript app.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*
- Built an AI-powered document platform for semantic search, summarization, and conversational QA over 50K+ records, combining vector retrieval, structured storage, and collaborative knowledge workflows.

HMIS Live | *MERN, TypeScript, ABDM APIs, Twilio, Sentry*
- Built a healthcare operations platform with dashboards, scheduling, and alerts, reducing administrative workload by 30% through TypeScript services, structured workflows, monitoring, and responsive views.